

# Prashant Sah

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## RESEARCH EXPERIENCE

### Research Assistant, Caldwell University

Sept 2026 - Present

Advisor: Prof. Bella Veksler | Project: *Do Large Language Models Know When Their Working Memory Fails?*

- Investigating LLM working-memory performance and metacognitive awareness using verbal and spatial n-back tasks.
- Reviewing cognitive science and machine learning research to identify limitations in existing LLM working-memory evaluations.
- Designing experiments to evaluate accuracy, confidence calibration, and performance across increasing memory demands.
- Developing an evaluation framework using Expected Calibration Error (ECE) and Brier Score.

### Undergraduate Research Assistant, Caldwell University

Sept 2025 - Aug 2026

Advisor: Prof. Vlad Veksler | Project: *Scaling Long-Range Retrieval: A Comparison of LSTM, Transformer, and Mamba on a Needle-in-a-Haystack Task*

- Developed a synthetic Needle-in-a-Haystack task to compare long-range retrieval abilities of LSTM, Transformer, and Mamba models.
- Evaluated model performance across sequence lengths from 32 to 1,024 tokens using PyTorch.
- Observed Transformer retrieval accuracy decline from approximately 99% at shorter sequences to 19% at 1,024 tokens.
- Compared retrieval accuracy, training time, and memory usage to analyze performance and computational efficiency.
- Poster accepted at CCSC Eastern Conference 2026, scheduled for October 9-10, 2026.

## PRESENTATIONS

### CCSC Eastern's 42nd Annual Regional Conference

Oct 2026

*Scaling Long-Range Retrieval: A Comparison of LSTM, Transformer, and Mamba on a Needle-in-a-Haystack Task*

### Caldwell University Annual Research and Creative Arts Day

May 2026

*Scaling Long-Range Retrieval: A Comparison of LSTM, Transformer, and Mamba on a Needle-in-a-Haystack Task*

## WORK AND LEADERSHIP EXPERIENCE

### Data Analyst Intern, Office of Institutional Research, Caldwell University

May 2026 - Aug 2026

- Built interactive dashboards and data visualizations to communicate research findings to faculty, staff, and administration.
- Developed and launched surveys using the IPEDS database, including data collection, validation, and analysis.
- Cleaned, organized, and analyzed institutional datasets using R.
- Conducted statistical analysis and data mining to extract actionable insights from complex datasets.

### NJAS Judging Website, Maintenance and Feature Update

Apr 2026

- Took over the New Jersey Academy of Science judging website after the original developer became unavailable.
- Partnered with three peers and the NJAS organizing team to deliver required changes ahead of the competition cycle.
- Modified site functionality to align with updated judging criteria and user-flow requirements, coordinating with stakeholders to validate scope.
- Debugged and patched existing code to ensure reliable performance during active judging and documented changes for future handoffs.

### Teaching Assistant, CS 196: Computer Programming II, Caldwell University

Jan 2025 - May 2025

- Assisted with assignment design and grading coursework for 50+ students in Python programming, OOP, recursion, and Python libraries.
- Led weekly tutoring sessions to help students prepare for assignments, quizzes, and exams.
- Provided hands-on lab guidance, helping students debug code and develop problem-solving skills.

### Teaching Assistant, CS 302: Design and Analysis of Algorithms, Caldwell University

Aug 2025 - Dec 2025

- Provided academic support to 15 students in algorithm design, graph theory, complexity analysis, debugging, and code optimization.
- Guided learners through divide-and-conquer, dynamic programming, greedy algorithms, and graph traversal.
- Reviewed code submissions to improve efficiency, correctness, and adherence to asymptotic complexity targets.

### Student Technician, IT Department, Caldwell University

July 2024 - Aug 2025

- Supported campus-wide IT systems for students, faculty, and staff.
- Troubleshoot university portal access, account issues, and hardware/software problems.
- Rebuilt and configured computers to improve workstation performance and reliability.

### Media Assistant, Caldwell University

Jan 2024 - Aug 2025

- Managed audio/visual setup, operation, and teardown for campus events, lectures, and livestreams.
- Operated cameras, microphones, mixing boards, and video-conferencing systems; troubleshoot issues during live events.
- Produced and edited multimedia content in support of university communications.

- Peer Tutor, Academic Success Center, Caldwell University

Sept 2024 - Present

  - Tutored 25+ students in Python and algorithms, strengthening computational problem-solving skills.
  - Guided learners through web development, discrete math, and algorithmic logic.
  - Helped students apply programming concepts to data analysis tasks and efficient code design.
- Campus Store Team Member, Barnes & Noble, Caldwell University

July 2024 - May 2026

  - Analyzed textbook inventory, enrollment data, and sales trends to forecast demand and maintain accurate stock levels.
  - Used POS and inventory systems to ensure accurate data entry and support data-driven operational decisions.

PROJECTS

- Next Word Predictor | PyTorch, LSTM, NLTK | GitHub

  - Designed and implemented an LSTM-based next-word prediction model using PyTorch with an embedding, LSTM, and linear architecture.
  - Built an NLP pipeline with NLTK tokenization, vocabulary construction (249 tokens), sequence generation (631 sequences), and padded batching.
  - Trained for 50 epochs using Adam optimizer and cross-entropy loss, achieving 99.52% next-word prediction accuracy.
  - Implemented single-step and iterative text generation for autocomplete-style sequence completion.
- Video Content Retrieval-Augmented Generation (RAG) System | GitHub

  - Engineered end-to-end machine learning pipelines using Python, pandas, and scikit-learn for preprocessing, feature extraction, and model training.
  - Implemented NLP workflows integrating OpenAI Whisper and BGE-M3 embeddings for semantic text analysis and context understanding.
  - Developed RAG models using vector embeddings and cosine similarity for context-aware information retrieval.
  - Automated ingestion, transformation, and parallel processing for JSON and multimedia datasets using pandas, NumPy, and Joblib.

EDUCATION

- Caldwell University, Caldwell, NJ

May 2027

Major: Computer Science | Double Minor: Mathematics, Business Analytics  
Dean's List, Honors Award Recipient, Presidential Scholarship  
Relevant coursework: Data Structures and Algorithms, Web Development, Database Management, Calculus, Artificial Intelligence

SKILLS

- Languages:

Python, R, JavaScript, C, SQL (MySQL), HTML/CSS
- Libraries and Frameworks:

PyTorch, TensorFlow, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Flask, NLTK, Shiny
- Tools:

Git/GitHub, Jupyter Notebook, VS Code, Excel, Word
- Domains:

Machine Learning, Deep Learning, NLP, Data Analysis, Data Visualization, Research